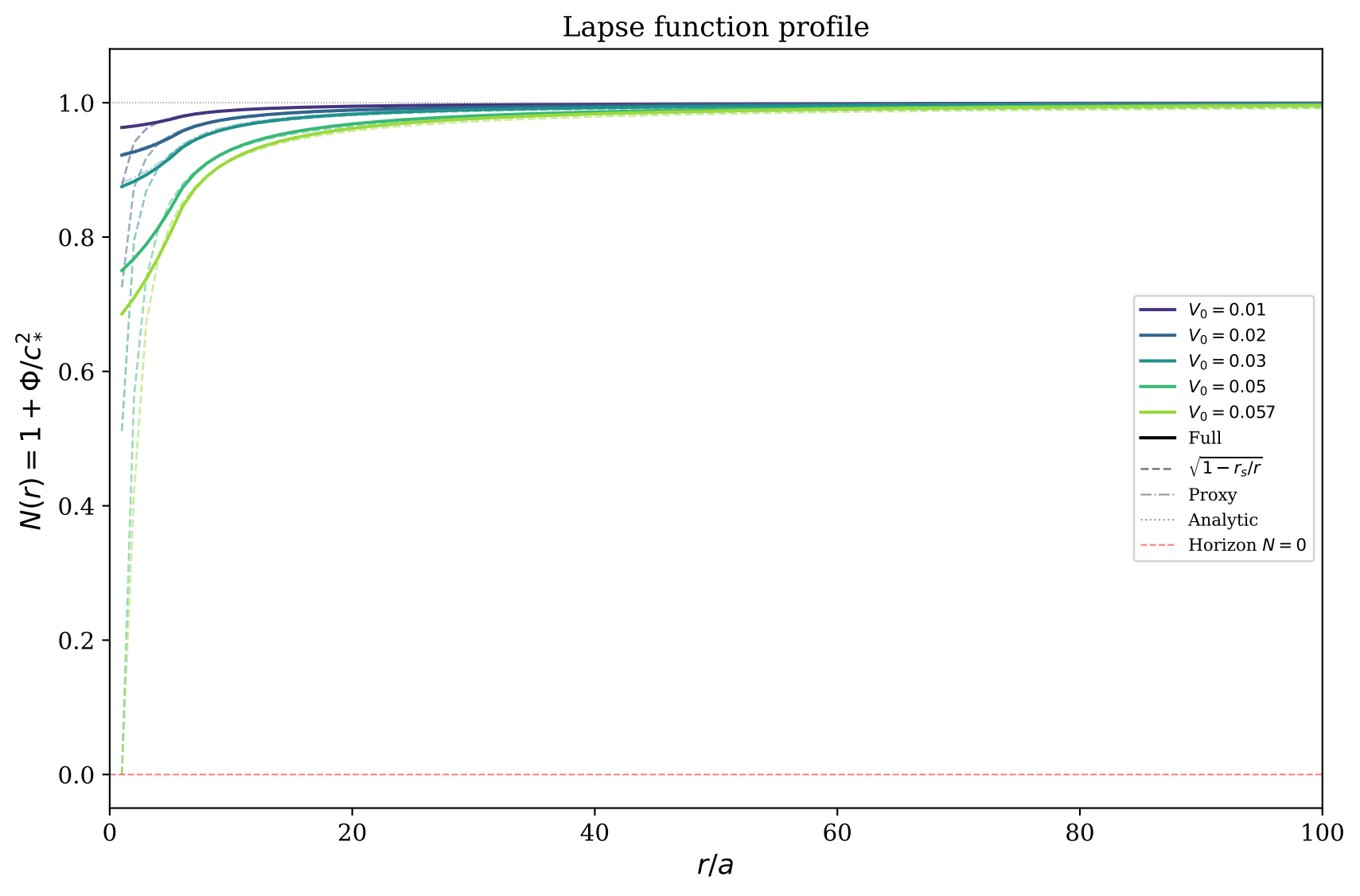
Lapse function profile
$N(r) = 1 + \Phi/c^2_*$
r/a
1.0
0.8
0.6
0.4
0.2
0.0
0
20
40
60
80
100
$V_0 = 0.01$
$V_0 = 0.02$
$V_0 = 0.03$
$V_0 = 0.05$
$V_0 = 0.057$
Full
$\sqrt{1 - r_s/r}$
Proxy
Analytic
Horizon $N = 0$